

PM & PMO Role — Requirements Document

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1. Overview

This document defines the requirements for introducing two new roles into SimERP: Project Manager (PM) and Project Management Office (PMO). These roles introduce a project-portfolio-based access model alongside a structured allocation ticketing system. The system is ticket-driven by design: with the single exception of PMO direct allocation (§6.3), every allocation change flows through a ticket to reduce manual intervention and preserve traceability.

PM assignment to projects is managed externally via the Zoho Analytics API. SimERP syncs project-to-PM mappings from Zoho — there is no manual PM assignment UI within SimERP.

2. Background

Existing roles (unchanged)

Role	Visibility
Employee	Own profile only
Reporting Manager (RM)	Their team of 10–12 employees
Team Manager (TM)	All employees in their department
Engineering Manager (EM)	All employees across their departments
Admin	Full system access

3. New Roles

3.1 Project Manager (PM)

A PM is responsible for managing one or more projects. Their visibility in SimERP is scoped to the projects they manage — they do not access department or team data.

Key characteristics:

- Each project has exactly one PM, sourced from the Zoho Analytics API sync
- A PM can manage multiple projects simultaneously
- PM can coexist with an org-hierarchy role (RM, TM, or EM) on the same employee — see §3.3
- PM only has access to two pages: PM Projects and Requests
- PMs cannot directly modify allocation records and cannot edit allocations inline — every change must be raised as a request (ticket)
- Tickets belong to the project, not the PM. If the PM on a project changes (via Zoho sync), the new PM inherits all existing tickets for that project; the previous PM loses visibility

3.2 Project Management Office (PMO)

The PMO coordinates between PMs (who request developers) and the approvers — TMs and EMs (who supply them). They have organisation-wide visibility and own the allocation workflow.

Key characteristics:

- Sees all projects and all employees across the organisation
- Owns the allocation ticketing workflow — triages tickets, assigns departments, shortlists developers, drives approvals, and acknowledges de-allocations
- Exclusive direct allocation: only PMO holds the “Add Allocation” permission to create a ProjectAssignment directly without a ticket (§6.3). PMs, TMs, and RMs are explicitly denied this action to maintain system integrity. Each direct allocation triggers an in-app + email notification to the project’s PM, the department’s TM/EM, the allocated employee’s RM, and the allocated employee
- PMO is a standalone role — cannot be combined with RM, TM, or EM
- Has full access within the PM_PROJECTS and ALLOCATION_TICKETS modules.
- Acts as L1 support for PMs navigating the ticket types (operational note)

3.3 Dual Role: PM combined with any org role

PM can be combined with any org-hierarchy role (RM, TM, EM). The rules are consistent regardless of which org role is paired:

Context	Active role
PM Projects page	PM scope — project-based visibility only
Requests page	Both views via tabs when paired with an approver role (TM or EM) — see §4.2
All other pages (employees, teams, departments, etc.)	Org role scope — broadest org role applies (EM > TM > RM > Employee). PM scope has no effect here.

Examples:

- PM + TM accessing the employees page → sees employees in their department (TM scope), not project employees
- PM + EM accessing the employees page → sees employees across all their departments (EM scope)
- PM + RM accessing the employees page → sees their team only (RM scope)

This means PM-scoped access is contained to the two PM pages only. On all other pages, the org role governs data visibility as if the PM role did not exist.

Approver pairing: where the paired org role is an approver (TM or EM), the Requests page also surfaces that role’s department approval queue (see §4.2 and §7). RM has no approval queue, so PM + RM shows the single PM Requests table only.

4. PM Experience**4.1 PM Projects Page**

Shows a table of the projects the PM is currently managing. Active and Completed/Archived projects are shown separately. If the PM on a project is changed, how that old project’s data is shown to the PM is deferred to the next sprint — see §12 Out of Scope.

Search & display:

- Search by project name
- Sort by: project name (A–Z), last updated date

- Pagination: 100 / 200 / 300 items per page
- Summary bar above the table showing total active projects count

Clicking a project opens a detail view with:

- List of all developers currently allocated to that project
- Per developer: name, job title, department, allocation type, allocation %, start/end date, **hours logged on this project (sourced from Zoho Analytics)**, leave details (upcoming, current, and past leaves), notice period status (if serving notice)

Developer list display:

- Search by developer name
- Sort by: developer name, allocation %, start date
- Filter by: leave status, notice period
- No pagination (developer lists per project are expected to be small)

Per-developer actions:

Action	Applies to	Description
Update Allocation	Fixed allocations	Opens a panel to modify allocation terms → creates an Update Allocation (UA) ticket. Decreases follow the lighter acknowledgement flow (§5.2.2); increases follow the full approval flow. Setting New Allocation Percentage to 0, or any relative reduction $\geq 70\%$, surfaces the ratings, feedback, and de-allocation reason fields.
Update Allocation	Dynamic allocations	The percentage field is disabled for dynamic allocations. The Update Allocation action raises a UA ticket that internally sets the allocation to 0% and the allocation end date to the current date; on apply, the ProjectAssignment is archived (soft-deleted). The ticket follows the lighter acknowledgement flow (§5.2.2). As a full de-allocation, it surfaces the ratings, feedback, and reason fields.

Dynamic allocation expectation note: wherever a dynamic allocation is displayed or requested, the system shows a note clarifying that dynamic allocation implies a maximum availability of 1–2 hours per day or 10–20 hours per month. If the PM needs a resource for a significant spike in work (e.g. 200 hours in a month), they must raise a new Fixed NA ticket — a dynamic allocation cannot be converted via UA.

Note: Developer Review and Developer/Project Escalation actions are deferred to a future sprint — see §12.

4.2 Requests Page

The ticketing dashboard. Shows all tickets associated with projects managed by this PM (tickets are project-owned, not PM-owned).

Page layout:

- If employee has PM role only (or PM + RM) → single table view
- If employee has PM plus an approver role (PM + TM or PM + EM) → two tabs:
 - “My Requests” — PM view (tickets for their projects)
 - “Department Requests” — approver view (tickets assigned to their department(s); the approval queue). A TM sees their single department; an EM sees all departments they manage.

Summary bar (above the table, for each tab):

- Count of tickets by state: Open / In Review / Awaiting Confirmation / Approved / Closed

Table columns: Ticket ID, Type (NA / UA), Project, Developer (if applicable), Tech Stack, State, Created Date, Last Updated

Search & filter:

- Search by ticket ID or project name
- Filter by: state (multi-select), type (NA / UA), project (dropdown), created date range
- Sort by: created date descending (default), last updated, ticket ID
- Pagination: 100 / 200 / 300 items per page

Create Request button → modal with three options: New Allocation (NA), Update Allocation (UA), and Replacement. The user sees three request types in the form, but the system stores only two ticket types (NA and UA) — a Replacement is saved as a UA ticket with a replacement flag (§4.2.3).

4.2.1 New Allocation (NA)

Used when a PM needs a new developer added to a project.

One form submission can include multiple resource requests. Each requested developer creates one separate ticket with its own ticket ID and state.

Fields per resource request:

Field	Required	Notes
Project	Yes	Dropdown — PM's active projects only
Primary Tech Stack	Yes	Dropdown — skills from SimERP tech stack list
Required Tech Stacks	No	Multi-select from SimERP tech stack list
Start Date	Yes	Cannot be in the past
End Date	No	Must be after start date if provided
Allocation Type	Yes	Fixed / Dynamic. Selecting Dynamic displays the expectation note (§4.1)
Allocation Percentage	If Fixed	Integer 1–100. Not applicable for Dynamic. (Show the percentage converted into hours as info next to the percentage field. Example: If the user selects 50% allocation show '4 hrs/day' next to it. This is to be done both in case of New allocation and update allocation ticket)
Developer Preference	No	Dropdown — any active employee. PM's preferred candidate; PMO may or may not honour it
Preferred Seniority	No	Junior / Mid / Senior / Lead
Request Type	Yes	Fresh Allocation / Shadow Developer / PR Reviewer / Estimation
PR Review Frequency	If PR Reviewer Selected	Dropdown options, considering frequency of PRs per week [Frequently, Occasionally, Rarely, Not at all]

Field	Required	Notes
Shadow For	If Shadow Developer	Dropdown — developers currently on the selected project
Interview Required	Yes	Yes / No (informational flag for PMO/TM/EM during review; no separate lifecycle step)
Client Facing	Yes	Yes / No

Request type notes:

- Fresh Allocation — Standard allocation of a developer to the project
- Shadow Developer — A developer who will shadow an existing team member. The “Shadow For” field is required.
- PR Reviewer — A developer allocated specifically to review pull requests. Follows the standard ticket lifecycle (no bypass). On approval, the developer is added as an “other allocation” PR reviewer on the project (§5.3); PR reviewers are surfaced on project allocation views via a marker column (§11 — Request Type persistence).
- Estimation — A short-term allocation (hours to a few days) for a developer to provide effort estimation for a project or feature. End Date is optional. Follows the standard ticket lifecycle. On approval, creates a short-duration ProjectAssignment like any other allocation.

4.2.2 Update Allocation (UA)

Used to modify or end an existing developer’s allocation on a project. Covers partial changes (update terms) and full deallocation (end engagement). Decreases follow the lighter acknowledgement flow; increases follow the full approval flow (§5.2).

Fields:

Field	Required	Notes
Project	Yes	Dropdown — PM’s active projects only
Developer	Yes	Dropdown — developers currently allocated to the selected project (Except for PM)
New Allocation Type	No	Fixed / Dynamic. Will auto populate as per current assignment, Leave as it is to keep current value. A dynamic allocation cannot be converted to Fixed via UA — raise a new Fixed NA instead (§4.1)
New Allocation Percentage	Yes (Fixed allocations)	Integer 0–100. 0 = full deallocation. Disabled for Dynamic allocations — the Update Allocation action sets 0% internally (§4.1)
New Start Date	No	Leave blank to keep current value
New End Date	No	Leave blank to keep current value

If New Allocation Percentage = 0, or the relative reduction is $\geq 70\%$ (relative reduction = $(\text{current \%} - \text{new \%}) \div \text{current \%}$; e.g. $100\% \rightarrow 30\%$, or $50\% \rightarrow 15\%$), these additional fields appear and are mandatory:

Field	Required	Notes
De-allocation Reason	Yes	Dropdown — Project Closure / Performance Issues / Role Misalignment / Downsizing
Core Skill Rating	Yes	Integer 1–10
Communication Rating	Yes	Integer 1–10
Attitude Rating	Yes	Integer 1–10
Justice to Role Rating	Yes	Integer 1–10
Developer Feedback	No	Free-text
Open to Work With Again	Yes	Yes / No

For Rating:

- The developer ratings added at time of deallocation should only be visible to RM, TM, EM, Admin, PMO.
- The PM can only see the previous rating that they have added for the developer.
- The bypass approval checkbox should only be visible to 'PMO'
- Note: The placement of the ratings are not visible anywhere else yet, so for now it will be visible in tickets only

4.2.3 Replacement

Replacement is presented to the user as a third option in the Create Request form. Internally it is stored as an Update Allocation (UA) ticket with a replacement flag, and the related replacement information is saved on that ticket — the system recognises only two ticket types (NA and UA).

For a Replacement request, the allocation data is pre-filled from the existing assignment being replaced, and the following replacement fields are additionally available:

Field	Required	Notes
Replacement Of	Yes	Reference to the developer/assignment being replaced
Replacement Type	Yes	Permanent / Temporary
Replacement Reason	Yes	Select Options: [Performance Issues, Role Misalignment, Long leaves, Allocation optimisation]

Replacement scope safeguard: replacement requests pre-populate the original developer's department and tech stack as non-editable fields to ensure a like-for-like replacement. A note on the form states that changing the technology stack (a "scope swap", e.g. replacing a QA with a Node.js developer) is not permitted through the replacement flow — it requires a fresh de-allocation plus a new standard NA ticket.

5. Ticket Lifecycle

5.1 Ticket Types

Type	Enum value	Ticket ID prefix	Purpose
New Allocation	NEW_ALLOCATION	NA-	Request a new developer for a project
Update Allocation	UPDATE_ALLOCATION	UA-	Modify or end an existing developer's allocation. Replacement requests are stored as UA tickets with a replacement flag (§4.2.3)

Ticket numbers are zero-padded 4-digit sequential integers per type (e.g., NA-0001, UA-0042). The sequence does not reset yearly. Stored internally as a ticketNumber Int column; the display string (e.g. NA-0001) is computed in the service layer by prepending the type prefix. Two PostgreSQL sequences (na_ticket_seq, ua_ticket_seq) provide atomic, gap-free numbering per type. Ticket detail URLs use the human-readable ID as the route parameter (e.g. /allocation-tickets/NA-0042) so URLs are shareable and meaningful to recipients.

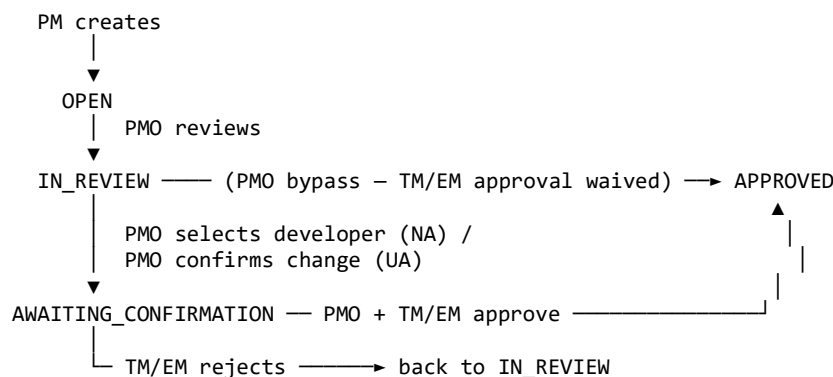
Tickets are never hard-deleted. Cancellation, archival, and any removal are implemented as soft delete (deletedAt) so the audit trail is preserved.

5.2 State Machines

Tickets follow one of two flows depending on the nature of the change. Both flows share the single AWAITING_CONFIRMATION state.

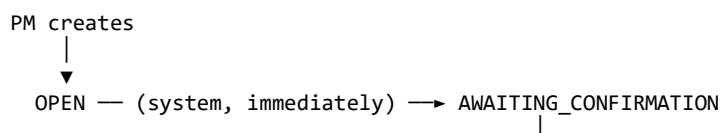
5.2.1 Full approval flow — NA tickets and UA increases

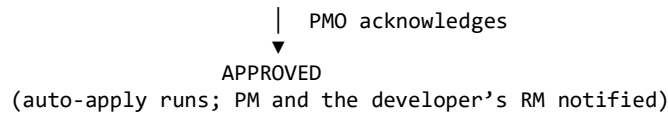
Applies to all NA tickets and UA tickets that increase the allocation percentage. Confirmation is two-party: one PMO representative AND the TM or EM of the assigned department must approve (the TM/EM approval is waived when PMO uses Bypass).



5.2.2 Lighter acknowledgement flow — UA decreases and de-allocations

Applies to any UA that decreases the allocation percentage — partial downgrades (e.g. 100% → 50%) and full de-allocations (0%), including dynamic-resource de-allocations raised via the Update Allocation action. De-allocations cannot be refused, so there is no review stage and no reject path.





Common rules:

- Closing a ticket → CLOSED. The PM can close a ticket when the allocation is no longer wanted, the project is cancelled, or for any other case where the ticket should not proceed. PMO can also close tickets for required control. Allowed from OPEN, IN_REVIEW, or AWAITING_CONFIRMATION.
- APPROVED tickets cannot be closed — once a ticket reaches the Approved stage, the allocation is done.
- To end an approved allocation, raise a new UA at 0% (for dynamic resources, the Update Allocation action sets 0% internally).

State	Meaning	Who acts
OPEN	Raised by PM, awaiting PMO	PMO
IN_REVIEW	Full flow only. NA: PMO assigned a department; PMO + TM/EM shortlisting candidates. UA increase: PMO review-and-confirm step only — no department selection (auto-inferred) and no shortlisting.	PMO + TM/EM
AWAITING_CONFIRMATION	Full flow: developer selected by PMO (NA) or change confirmed by PMO (UA increase); awaiting two-party approval — one PMO representative and the TM/EM of the assigned department (TM/EM approval waived on PMO bypass). Lighter flow: decrease/de-allocation awaiting PMO acknowledgement; cannot be rejected.	PMO + TM/EM
APPROVED	On Save, Approved, acknowledged, or PMO-bypassed; allocation change auto-applied	System
CLOSED	Closed by PM (ticket not proceeding) or by PMO (required control) before fulfilment (soft-deleted, retained for audit)	PM / PMO

5.3 On APPROVED — Automatic Allocation Change

Whether a ticket reaches APPROVED via the two-party approval, PMO bypass, or PMO acknowledgement, the system automatically applies the change in a transaction:

- NA approved → creates a new ProjectAssignment with status = GREEN and typeOfWork = DEVELOPMENT_WORK for any type of allocation. In case of 'PR Reviewer' type tickets, do not treat it as a project allocation — treat it as an 'other allocation' and add the developer as PR reviewer in that project. Other allocations count over the 100% allocation limit (they may exceed it). The originating requestType is persisted on the assignment.
- UA (UPDATE_ALLOCATION) approved/acknowledged with % > 0 → updates the existing ProjectAssignment (only the fields whose values were changed; unchanged pre-filled fields are left as is)

- UA (UPDATE_ALLOCATION) approved/acknowledged with % = 0 (including dynamic Update Allocation de-allocations) → sets allocationEndDate = approval/acknowledgement date, then soft-deletes (archives) the ProjectAssignment

Before applying, the system verifies the target ProjectAssignment still exists and deletedAt IS NULL. If the assignment was already deleted by another approved ticket, the transaction fails.

If the transaction fails, the ticket reverts to its prior state (AWAITING_CONFIRMATION) and PMO is notified with a retry prompt.

Validation

To move a ticket from the AWAITING_CONFIRMATION state to APPROVED, approvals are required from two parties:

- One PMO representative
- The TM/EM of the selected department

If PMO selects the Bypass checkbox, the TM/EM's approval is not required — the PMO approval alone is sufficient.

The ticket should clearly display which approvers have already provided approval and which approvals are still pending. If approval is bypassed by PMO then it should show the bypassed approver with separate indicator.

If any modifications are made regarding the assignment to the ticket after approval has been provided by either party, all existing approvals should be invalidated, and the ticket should automatically revert to the AWAITING_CONFIRMATION state.

5.4 Shortlist (NA tickets only, IN_REVIEW phase)

During IN_REVIEW of a NA ticket, PMO and the TM/EM collaboratively add candidate developers to a working shortlist. PMO then selects one final developer, moving the ticket to AWAITING_CONFIRMATION.

UA tickets have no shortlist — the developer is already identified via projectAssignmentId at creation time.

The shortlist is not visible to the PM. The PM only sees the ticket state.

The shortlist is ephemeral — once a developer is selected and the ticket moves forward, the shortlist data is no longer meaningful.

Adding a developer to a shortlist shows a warning (but does not block) if that developer is already shortlisted on another active ticket.

5.5 Bypass Approval (full flow only)

PMO can bypass the TM/EM approval in emergency scenarios — when the Bypass checkbox is selected, the TM/EM's approval is not required and the PMO approval alone moves the ticket to APPROVED (directly from IN_REVIEW). The TM/EM is notified when their approval is bypassed. No reason is required from PMO, but the bypass is recorded on the ticket for audit purposes via bypassedAt (timestamp) and bypassedByld (FK to the PMO employee who triggered the bypass). Lighter-flow tickets have no approval to bypass.

5.6 Approver Rejection (full flow only)

The TM or EM of the assigned department can reject a ticket in AWAITING_CONFIRMATION. The ticket returns to IN_REVIEW for PMO to shortlist a different developer. No rejection reason is required. Lighter-flow (decrease) tickets cannot be rejected — de-allocations cannot be refused.

PM not notified: The PM is intentionally not notified of a rejection. A rejection is handled entirely between PMO and the approver (PMO re-shortlists and re-submits). The PM continues to see only the ticket state and will be notified when the allocation is ultimately approved.

6. PMO Experience

6.1 Requests Page

PMO sees all tickets from all PMs across the organisation.

Summary bar (above the table):

- Count of tickets by state: Open / In Review / Awaiting Confirmation / Approved / Closed

Search & filter:

- Search by ticket ID, project name, PM name, developer name
- Filter by: state (multi-select), type, department, tech stack/skill, PM (dropdown), created date range
- Sort by: created date descending (default), last updated, state, department
- Pagination: 100 / 200 / 300 items per page

Pending Actions section: a dedicated PMO view aggregating actionable alerts across the system — tickets awaiting acknowledgement, upcoming de-allocation reminders, and failed auto-applies awaiting retry — so PMO can monitor and resolve them centrally.

6.2 Ticket Detail View

On clicking a ticket, PMO can:

For NA tickets:

- Assign Department — Select which department should supply the developer. PMO uses the primary tech stack and required tech stack fields along with their knowledge of departmental availability to decide. Assignment moves ticket to IN_REVIEW.
- Shortlist Developers — Add candidate developers from the assigned department. The TM/EM can also add to this list. A warning is shown (not a block) if an added developer is already shortlisted on another active ticket.
- Select Final Developer — Pick one developer from the shortlist → ticket moves to AWAITING_CONFIRMATION → TM/EM notified.
- Bypass Approval — Checkbox for emergencies. The TM/EM approval is waived; the ticket moves directly from IN_REVIEW to APPROVED. The TM/EM is notified of the bypass.

For UA increase tickets (full flow):

- Review and Advance — PMO reviews the requested change and moves the ticket from OPEN to IN_REVIEW. assignedDepartmentId is set automatically from the target developer's department — no manual selection.
- Send for Approval — PMO confirms the change is appropriate and moves the ticket from IN_REVIEW to AWAITING_CONFIRMATION → TM/EM of the developer's department notified.
- Bypass Approval — Checkbox for emergencies. The TM/EM approval is waived; the ticket moves directly from IN_REVIEW to APPROVED. The TM/EM is notified of the bypass.

For AWAITING_CONFIRMATION tickets (full flow):

- Approve — Records the PMO approval. The ticket moves to APPROVED once both the PMO and TM/EM approvals are present (the TM/EM approval is waived if Bypass was used). The ticket displays which approvals are recorded and which are pending.

For UA decrease / de-allocation tickets (lighter flow):

- Acknowledge — Moves the ticket from AWAITING_CONFIRMATION to APPROVED. The allocation change auto-applies (§5.3) and the PM and the developer's RM are notified that the change is finalised. There is no reject option.

6.3 Direct Allocation (Add Allocation)

PMO holds an exclusive “Add Allocation” action to create a ProjectAssignment directly on any project without raising a ticket. This is the only non-ticket allocation path in the system.

- Available to PMO only — PMs, TMs, and RMs are explicitly denied (enforced via a dedicated CASL permission on the PMO role)
- The creating PMO employee is recorded (createdById) for traceability
- On creation, an in-app + email notification is sent to the project's PM, the department's TM/EM, the allocated employee's RM, and the allocated employee

7. Approver Experience (TM / EM)

Approvers — TMs and EMs — gain a Requests section (new tab or page in their navigation). A TM approves for their single department; an EM approves for all departments they manage. Approver queues contain full-flow tickets only — lighter-flow (decrease) tickets are acknowledged by PMO and do not require TM/EM action.

7.1 Department Requests Tab

Shows all tickets assigned to the approver's department(s).

Summary bar — count by state: Open / In Review / Awaiting Confirmation / Approved / Closed

Search & filter:

- Search by ticket ID, project name, PM name
- Filter by: state (multi-select), type, created date range (EM additionally: department)
- Sort by: created date descending (default), last updated
- Pagination: 100 / 200 / 300 items per page

Sub-view: Pending My Approval — filtered view of AWAITING_CONFIRMATION tickets awaiting the TM/EM approval (for quick access to action items).

7.2 Actions on AWAITING_CONFIRMATION Tickets (full flow)

- Approve → records the TM/EM approval. The ticket moves to APPROVED once the PMO approval is also present (the TM/EM approval is only waived by PMO bypass); the allocation change is then auto-applied and the PM notified
- Reject → ticket returns to IN_REVIEW, PMO notified. No reason required. (PM is not notified — see §5.6.)

7.3 Notifications for Approvers (TM / EM)

Event	Notification
Ticket assigned to their department	In-app + email
Developer selected, pending their approval	In-app + email
Their approval bypassed by PMO	In-app + email

8. Notifications and Alerts

Alerts:

- On click on 'save' / 'approve' of any ticket type, user should see a confirmation dialog asking are you sure you want to save the changes

Notifications:

All notifications are delivered via in-app (bell icon) + email. Integration with the notification module being built in parallel. PMO additionally sees actionable alerts aggregated in the Pending Actions section (§6.1).

General rule: in every allocation — direct (§6.3) or ticket-based — the allocated employee and the employee's RM receive the notification along with the other recipients listed below.

Event (trigger)	Recipient(s)	Channel
Ticket created (NA or UA) — fired on PM submit; one per generated ticket	PMO	In-app + email
Ticket assigned to department (→ IN_REVIEW)	TM / EM of the assigned department	In-app + email
Developer selected, pending approval (→ AWAITING_CONFIRMATION)	TM / EM of the assigned department	In-app + email
Ticket approved — full flow (→ APPROVED)	Current PM of the project; and the allocated developer, RM of developer, PMO, TM of department (message includes project name and PM name)	In-app + email
Decrease ticket acknowledged — lighter flow (→ APPROVED)	Current PM of the project; the developer's RM; and the developer (confirms the change is finalised)	In-app + email
Ticket rejected by approver (→ IN_REVIEW)	PMO only (PM is not notified — see §5.6)	In-app + email
Approval bypassed by PMO (→ APPROVED)	TM / EM whose approval was bypassed; and current PM	In-app + email
De-allocation reminder — 5 days before scheduled allocation end date	PM (also surfaced in PMO Pending Actions)	In-app + email
De-allocation reminder — 1 day before scheduled allocation end date	PM (also surfaced in PMO Pending Actions)	In-app + email
Direct allocation created by PMO (§6.3)	Project's PM + department's TM/EM + allocated employee's RM + allocated employee	In-app + email
Ticket closed by PM	PMO notified in case of any status except for Open	In-app + email

9. Searching, Sorting, Filtering & Pagination

All tables use the standard pagination limits: 100 / 200 / 300 items per page (selectable).

Page / Table	Search	Default Sort	Available Filters
PM Projects (project list)	Project name	Project name A–Z	None
PM Projects (developer list)	Developer name	Developer name A–Z	Leave status, Notice period
PM Requests	Ticket ID, project name	Created date desc	State, Type (NA/UA), Project, Date range
PMO Requests	Ticket ID, PM name, project, developer	Created date desc	State, Type (NA/UA), Department, Skill, PM, Date range
Approver Requests (TM/EM)	Ticket ID, PM name, project name	Created date desc	State, Type (NA/UA), Date range (EM: + Department)
PMO Pending Actions	Project name, developer name	Due date asc	Alert type (acknowledgement / reminder / failed auto-apply)

10. Validation Rules

Field	Rule
Allocation % (NA ticket)	Integer 1–100 (Fixed only; not applicable for Dynamic)
Allocation % (UA ticket)	Integer 0–100 (0 = full deallocation). Disabled for Dynamic allocations — the Update Allocation action sets 0% internally
Relative reduction	$(\text{current \%} - \text{new \%}) \div \text{current \%}$. Threshold for mandatory ratings/feedback/reason is $\geq 70\%$
De-allocation Reason	Required when new % = 0 or relative reduction $\geq 70\%$. Values: Project Closure / Performance Issues / Role Misalignment / Downsizing
Ratings (de-allocation / $\geq 70\%$ reduction)	Integer 1–10; required when new % = 0 or relative reduction $\geq 70\%$
Ticket number	Zero-padded 4-digit sequential per type (NA-0001, UA-0001). No yearly reset.
Seniority values	JUNIOR / MID / SENIOR / LEAD
Start date	Cannot be in the past ($\geq$ today)
End date	Must be after start date if provided
Developer preference	Any active employee
Allocation % (FIXED type)	Required if allocation type is Fixed; not applicable for Dynamic
Replacement Type	PERMANENT / TEMPORARY; required on Replacement requests

Field	Rule
Replacement Reason	Required on Replacement requests. Values: Performance Issues / Role Misalignment / Long leaves / Allocation optimisation
Replacement department & tech stack	Pre-populated from the original developer; non-editable on replacement requests

11. System Design Decisions

These decisions were made during the design sessions (including Sprint 4 planning, 2026-06-03). Flag any disagreements before development begins.

Decision	Detail
Partial update semantics	UA forms are pre-filled with the current assignment values; the user updates only the fields requiring change. On apply, only fields whose values differ from the current assignment are updated — unchanged pre-filled fields are left as is. For Replacement requests, allocation data is pre-filled and replacement-specific fields are additionally shown (§4.2.3).
Partial deallocation	Updates ProjectAssignment.allocationPercentage in place (Option A). Full history is preserved via the ticket record itself.
Auto-apply failure	Ticket reverts to its prior state (AWAITING_CONFIRMATION). PMO is notified with a retry prompt. Failure is logged.
Concurrent shortlist writes	PostgreSQL atomic array operations handle concurrent PMO + TM/EM writes. Last-write-wins is acceptable for this low-frequency collaborative flow.
Duplicate UA tickets	If a PM attempts to create a UA for an employee+project combination that already has an active ticket (state ∈ OPEN, IN_REVIEW, AWAITING_CONFIRMATION), the request is blocked with a conflict error. For NA tickets (no specific developer at creation), no block applies. A shortlist-level warning is shown to PMO/TM/EM when adding a developer already shortlisted on another active ticket — this does not block the addition.
CASL module names	Two new permission modules: PM_PROJECTS (PM Projects page) and ALLOCATION_TICKETS (ticketing system). A dedicated PMO-only permission governs direct allocation (§6.3).
PM assignment	No manual assignment UI in SimERP. All project-to-PM mappings are synced from Zoho Analytics API nightly at 12 AM using a new PROJECT_METADATA sync job. Zoho provides projectManagerEmail; the sync matches it to Employee.email to set Project.projectManagerId (FK). Unmatched emails leave projectManagerId as null and log a warning. PM changes in Zoho take effect in SimERP the following day (up to 24-hour lag is acceptable).
Project.projectManager field	The existing projectManager String? free-text field is replaced by projectManagerId BigInt? (FK → Employee) in the same schema

Decision	Detail
	migration. The old string field is dropped; the Zoho sync repopulates the FK on its next nightly run.
Ticket ID storage	ticketNumber Int is stored on the AllocationTicket model. The display string (e.g. NA-0001) is computed in the service layer by prepending the type prefix. Two PostgreSQL sequences (na_ticket_seq, ua_ticket_seq) provide atomic generation per type. Ticket detail URLs use the human-readable ID as the route parameter (e.g. GET /allocation-tickets/NA-0042) so links are shareable. CASL still governs visibility — a shared URL returns 403 if the recipient lacks access.
UA → ProjectAssignment link	UA tickets store projectAssignmentId (FK → ProjectAssignment) at creation time to pin the exact assignment being modified. NA tickets leave this null. Before auto-apply, the system verifies the assignment still exists (deletedAt IS NULL); if already deleted the transaction fails and falls into the standard auto-apply failure path.
Deallocation ratings storage	Core skill, communication, attitude, justice-to-role ratings, developer feedback, and de-allocation reason stored as inline nullable columns on AllocationTicket. Populated when newAllocationPercentage = 0 or relative reduction ≥ 70%.
Ratings visibility	Deallocation ratings are visible only to RM, TM, EM, Admin, and PMO. A PM can only see the ratings they themselves previously added for the developer. Ratings are surfaced in tickets only for now (no other placement yet).
Bypass audit fields	bypassedAt DateTime? and bypassedById BigInt? (FK → Employee) stored on AllocationTicket. Both null for non-bypassed tickets. The Bypass checkbox is visible to PMO only.
Additional skills storage	additionalSkillIds BigInt[] (PostgreSQL array) on AllocationTicket. Primary tech stack stored as techStackSkillId BigInt (FK → Skill).
Active project filter	PM Projects page filters by Project.status using an ACTIVE_PROJECT_STATUSES constant. Exact Zoho status string values to be confirmed when the Zoho project-status API integration is finalised.
Auto-created assignment defaults	ProjectAssignment records created on ticket approval default to status = GREEN and typeOfWork = DEVELOPMENT_WORK. The originating requestType IS persisted on the assignment so PR Reviewers, Shadows, and Estimation allocations remain identifiable downstream. Exception: PR Reviewer tickets do not create a project allocation — the developer is added as an “other allocation” PR reviewer on the project, and other allocations count over (may exceed) the 100% allocation limit.
Request Type surfacing (PR Reviewers)	A marker column (working name “Other Locations”) on project allocation views identifies PR Reviewers brought onto a project. Exact placement and labelling decided during frontend design.
Deallocation end date	On 0% UPDATE_ALLOCATION approval/acknowledgement, allocationEndDate is set to the approval date before the soft-delete.

Decision	Detail
	deletedAt marks the deletion event; allocationEndDate marks the business end-of-engagement.
UA department auto-inference	When PMO advances a UA ticket to IN_REVIEW, assignedDepartmentId is set automatically from the target developer's Employee.departmentId. No manual department selection is required.
Approval model	Full-flow tickets in AWAITING_CONFIRMATION require two approvals: one PMO representative and the TM or EM of the assigned department (an EM may approve for any department they manage). The ticket displays which approvals are recorded and which are pending; a bypassed approver is shown with a separate indicator. PMO Bypass waives the TM/EM approval. If the ticket's assignment details are modified after either approval, all approvals are invalidated and the ticket reverts to AWAITING_CONFIRMATION. TM/EM rejection returns the ticket to IN_REVIEW.
Lighter flow routing	UA tickets that decrease allocation skip IN_REVIEW: on creation they move immediately (system transition) from OPEN to AWAITING_CONFIRMATION. New endpoint PATCH /allocation-tickets/:id/acknowledge (PMO only). No reject transition exists for this flow.
Ticket soft delete	deletedAt DateTime? on AllocationTicket. All ticket removal/cancellation is soft delete (archive) — never hard delete — to maintain audit logs.
Replacement storage	Replacement is a form-level request option stored as an UPDATE_ALLOCATION ticket with a replacement flag. replacementOfAssignmentId BigInt? (FK → ProjectAssignment, the engagement being replaced) and replacementType enum (PERMANENT / TEMPORARY) plus replacement reason are stored on the AllocationTicket and carried onto the resulting assignment to build replacement chains for future Allocation Accuracy analytics.
PMO direct allocation	Dedicated CASL permission on the PMO role only. createdById recorded on the assignment. In-app + email notification to the project's PM, the department's TM/EM, the allocated employee's RM, and the allocated employee on creation.
Allocation editing	PMs cannot edit allocations directly (no inline allocation editing); every change must be raised as a request (ticket). Direct allocation remains a PMO-only capability (§6.3).
Dynamic allocation note	System-displayed note: "Dynamic allocation implies a maximum availability of 1–2 hours per day or 10–20 hours per month." Significant spikes require a new Fixed NA ticket; dynamic allocations cannot be converted via UA.
Relative reduction formula	$(\text{currentPercentage} - \text{newPercentage}) \div \text{currentPercentage} \geq 0.70$ triggers mandatory ratings, feedback, and de-allocation reason.

Decision	Detail
State transition API design	Each ticket state transition is a separate endpoint (e.g. PATCH /allocation-tickets/NA-0001/assign-department, /shortlist, /select-developer, /bypass, /approve, /reject, /acknowledge, /close). Each has its own @CheckAbility decorator and typed request DTO.
Ticket ownership	Tickets belong to the project, not the PM. The current PM of a project (per latest Zoho sync) sees all tickets for that project.
Dual role scope isolation	PM scope applies only to PM_PROJECTS and ALLOCATION_TICKETS modules. On all other modules (EMPLOYEES, TEAMS, etc.), only the org role scope is active. The CASL permission system enforces this naturally since PM role holds no permissions for org-hierarchy modules.
Logged-hours source	“Hours logged on this project” shown on the PM Projects developer detail is sourced from Zoho Analytics, not computed within SimERP.

12. Out of Scope (This Sprint)

Item	Notes
Developer Review	Deferred — PM writing permanent qualitative records about developers
Developer Escalation	Deferred — PM flagging urgent issues with a specific developer
Project Escalation	Deferred — PM flagging broader project-level issues
Resignation & Leave Management	Deferred to a later sprint — auto-generated replacement tickets on resignation/LWD, temporary replacement handling for long leave, and return-from-leave alerts will be covered in later sprints.
Old project data after PM change	Active and Completed/Archived projects are shown separately. If the PM on a project is changed, the flow for how that old project's data is shown to the PM is pending for the next sprint.
PIP (Performance Improvement Plan) tracking	Not tracked in SimERP yet
Consolidated PM Rating on employee profile	Future feature — aggregate deallocation ratings across tickets
Allocation Accuracy reporting	Data capture (Replacement Of chains, Permanent/Temporary flags, de-allocation reasons) is in scope now. The mathematical reports and severity-based accuracy counts are future-facing analytics.
Bulk PM re-assignment	Not required — projectManager String? is dropped and replaced by projectManagerId BigInt? FK. The nightly Zoho sync repopulates all project-to-PM mappings automatically.

13. Glossary

Term	Definition
PM (Project Manager)	Employee responsible for managing one or more projects. Visibility scoped to their projects only. Assigned via Zoho sync.
PMO (Project Management Office)	Organisation-wide role responsible for allocation coordination. Sees all projects and employees. Sole holder of direct-allocation access.
Approver	The TM or EM of the assigned department who approves or rejects a full-flow ticket in AWAITING_CONFIRMATION. An EM may approve for any department they manage.
Allocation	A ProjectAssignment record linking an employee to a project for a defined period and percentage.
Fixed Allocation	Standard percentage-based allocation (e.g. 100%, 50%, 25%).
Dynamic Allocation	Need-based allocation (e.g. maintenance projects). Implies maximum availability of 1–2 hours/day or 10–20 hours/month. Percentage field is disabled; ended via the Update Allocation action, which sets the allocation to 0% internally.
Other Allocation	A non-project allocation that marks a developer as a PR reviewer on a project. Other allocations count over (may exceed) the 100% allocation limit.
New Allocation (NA)	Request to add a new developer to a project. Ticket ID prefix NA-, enum NEW_ALLOCATION.
Update Allocation (UA)	Request to modify or end an existing developer's allocation. Ticket ID prefix UA-, enum UPDATE_ALLOCATION. 0% = full deallocation. Replacement requests are stored as UA tickets with a replacement flag.
Awaiting Confirmation	Single confirmation state shared by both flows. Full flow: awaiting two-party approval (one PMO representative + the TM/EM; TM/EM waived on PMO bypass). Lighter flow: awaiting PMO acknowledgement; cannot be rejected.
Replacement Request	Presented as a third option in the Create Request form; stored as a UA ticket with a replacement flag, linked via Replacement Of. Permanent or Temporary. Department and tech stack are locked like-for-like.
Allocation Accuracy	Future analytics measuring the quality of initial allocations using replacement chains (e.g. repeated performance-driven replacements indicate low accuracy).
Pending Actions (PMO)	PMO dashboard section aggregating acknowledgements due, de-allocation reminders, and failed auto-applies.
Shortlist	The working list of candidate developers during IN_REVIEW of a NA ticket. Ephemeral and not visible to PM. Does not apply to UA tickets.
Developer Review	PM's permanent qualitative record about a developer on a project. Deferred to future sprint.

Term	Definition
Escalation	Active incident raised by PM requiring urgent intervention. Deferred to future sprint.